MILITARY SPECIFICATION

MOTOR-TACHOMETER GENERATOR, TEMPERATURE COMPENSATED

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. - This specification establishes the requirements for the procurement of one type of 400 cycle, temperature-compensated motor-tachometer generator, BUWEPS drawing 2104608.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, publications, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-A-355	Aluminum Alloy Plate and Sheet 2024
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

MILITARY

MIL-P-116	Preservation, Methods of

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MIL-S-901	Shockproof Equipment Class HI (High Impact) Shipboard Application, Tests for
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specifications for
MIL-F-18870	Fire Control, Optical and Associated Equipment Intended for Shipboard Use, the General Specification for Design and Procurement of

STANDARDS

MILITARY

MIL-STD-106	Mathematical Symbols
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment

DRAWINGS

NAVY DE PARTMENT

BUWEPS DWG 2104608	Motor-Tachometer Generator, Temperature Compensated

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

NAWEPS OS 6341	Miscellaneous General Ordnance Design Requirements

[Copies of specifications, standards, and drawings, requested by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other Publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATION

Handbook H28	Screw-Thread Standards for Federal Services

[Application for copies should be addressed to the Superintendent of Documents, Government Printing Office Washington 25, D. C.)

3. REQUIREMENTS

3.1 Description. - The motor-tachometer generator consists of a two-phase induction motor and a drag-cup induction generator which have a common shaft, and are housed to form an integral unit. The two-phase induction motor is of the servo type -- having approximately linear relationships between speed, control voltage, and torque. The control phase of the motor is divided into two separate windings which may be connected in series or in parallel. The drag-cup induction generator produces a 400-cycle voltage which is closely proportional to motor speed. The generator is temperature compensated by means of a heater and thermostat to reduce the effect of ambient air temperature on the output voltage.

3.2 General requirements.

3.2.1 Pre-production Samples. - Unless otherwise specified in the contract or order, pre-production samples of the motor-tachometer generators shall be manufactured using the methods and procedures proposed for the production lot. The sample shall be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's methods are capable of yielding items that comply with the technical requirements of the contract.

3.2.2 Materials. - Unless otherwise specified in the contract, requisition, or order, all materials entering into the final assembly of the motor-tachometer generator shall conform to the detailed requirements indicated on BUWEPS Dwg 2104608.

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3.2.3 Construction and Dimensions. - Constructions and dimensions for the motor-tachometer generator shall be in accordance with BUWEPS Dwg 2104608.

3.2.4 Design. - The product covered by this specification shall be capable of meeting the requirements of this document and BUWEPS Dwg 2104608.

3.2.5 Marking. - Marking of assemblies shall be in accordance with the applicable drawings.

3.2.6 Threads. - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings, and the NO-GO gaging practice shall be as set forth in the aforementioned Handbook H28 .

3.2.7 Requests for Deviation. - Each request that a deviation be authorized shall have been submitted to the Procuring Activity, and the deviation shall have been authorized by the Bureau of Naval Weapons or by a Naval Activity delegated such authority, prior to the production of the affected item, or prior to the performance of a particular operation on the item. The deviation shall apply only to this specific item or operation. The request shall define the suggested deviation in sufficient detail to permit the Procuring Activity to evaluate its effect on performance and reliability, and the reasons for the suggested deviation shall be stated. Each request that a deviation be authorized, and the authorization, shall state specifically the extent of the departure, the identifying number of the item, the contractor (or bidder) and the contract number. A deviation is essentially temporary in character, and usually represents a departure limited to a specific period, to a specific number of items, or to the contract to which it applies. It may require an equitable reduction in contract price if a decrease in the cost of performance results therefrom.

3.2.8 Conflicting Requirements, - Conflicting requirements arising between this specification or any specification, publication, or drawing listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

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3.3 Definitions. - Definitions applicable to this specification and associated drawings are as follows:

3.3.1 Ambient Temperature. - The temperature of the air surrounding the unit, measured along the shaft axis 1.5 plus or minus 1.0 inches away from the end of the unit opposite the shaft.

3,3.2 Starting Voltage. - The minimum control field winding voltage necessary to start and maintain smooth shaft rotation (with no mechanical load) with rated voltage of proper frequency and phase relationship applied to the motor fixed field and tachometer excitation-voltage winding.

3.3.3 No-Load Speed. - The shaft speed of rotation (with no mechanical load) with rated voltage of proper frequency and phase relationship applied to both motor windings and to the tachometer excitation-voltage winding.

3.3.4 Single Phasing. - The tendency of the rotor to continue to rotate when one motor winding is opened and the other motor winding remains energized.

3,3.5 Stall Torque. - The torque developed at speeds in excess of one revolution per minute but not less than one-half per cent of synchronous speed with rated voltage of proper frequency and phase re - lationship applied to both motor windings, and to the tachometer excitation-voltage winding.

3.3.6 Speed Range. - The range of speed, clockwise and counterclockwise, within which the specified performance parameters are applicable. The speed range is from zero to a specified speed (symbol N').

3.3.7 Calibration Speed. - A specified speed, counterclockwise and clockwise, within the speed range for purposes of output voltage standardization and reference (symbol No).

3.3.8 Tachometer output voltage.

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3.3.8.1 In-phase, quadrature, and harmonic components.

3.3.8.1.1 In-Phase Output Voltage. - That component of output voltage which is in phase (or 180 degrees out of phase) with the fundamental frequency component of tachometer excitation voltage. In-phase output voltage is positive if in-phase, and negative if 180 degrees out of phase.

3.3.8.1.2 Quadrature output Voltage. - That component of output voltage which leads (or lags) the fundamental frequency component of tachometer excitation voltage by 90 degrees. Quadrature output voltage is positive if leading, and negative if lagging.

3.3.8.1.3 Harmonic Output Voltage. - That component of output voltage which consists of frequencies other than the fundamental frequency of tachometer excitation voltage.

3.3.8,2 Zero-speed (null) output voltage.

3.3.8.2.1 Axis Voltage. - That component of zero-speed output voltage of fundamental frequency which is independent of rotor position.

3.3.8 .2.1.1 In-Phase Axis Voltage. - One-half the sum of the maximum and minimum in-phase zero speed voltages. See 3.3.8.1.1 and figure 4. In-phase axis voltage is also equal to one-half the sum of the in-phase output voltages at a given speed, counterclockwise and clockwise. See 3.3.8.1.1 and figure 4.

3.3.8 .2.1.2 Quadrature Axis Voltage. - One-half the sum of the maximum and minimum quadrature zero- speed voltages. See 3.3.8.1.2 and figure 5. Quadrature axis voltage is also equal to one-half the sum of the quadrature output voltages at a given speed, counterclockwise and clockwise.

3.3.8.2.2 Position Voltage. - That component of zero-speed output voltage of fundamental frequency which is a function of rotor position.

3.3.8 .2.2.1 In-Phase Position Voltage. - One half the difference of *the* maximum and minimum in-phase zero-speed voltages. See 3.3.8.1.1 and figure 4.

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3.3.8 .2.2.2 Quadrature Position Voltage. - One-half the difference of the maximum and minimum quadrature zero-speed voltages. See 3.3.8.1.2 and figure 5.

3.3.8.2.3 Harmonics. - The zero-speed harmonic output voltage. See 3.3.8.1.3.

3.3.8.3 Speed-Sensitive Output Voltage. - That component of output voltage of fundamental frequency which is a function of speed only.

3.3.8.3.1 In-Phase Speed-Sensitive Output Voltage. - One-half *the* difference of the in-phase output voltages counterclockwise and clockwise at the same rotational speed N (symbol E). E. is the special case of the in-phase speed-sensitive output voltage at calibration speed.

3.3.8.3.2 Quadrature Speed-Sensitive Output Voltage. - One-half the difference of the quadrature output voltages counterclockwise and clockwise at the same rotational speed N (symbol Q). Q. is the special case of the quadrature speed-sensitive output at calibration speed.

3.3.9 Output Voltage Gradient. - The in-phase speed-sensitive output voltage per unit speed. This parameter carries no tolerance.

3.3.10 Transformation Ratio. - The in-phase speed-sensitive output voltage per unit tachometer energizing voltage of fundamental frequency at a specified speed.

3.3.11 Phase Shift. - Phase shift is given by

$$\sim \; \text{‘}^{1Q} \text{for (identical conditions of test for E and Q)}$$

where E and Q are defined as in 3,3.8. 3.1 and 3.3.8.3.2. Mathematical symbols shall be interpreted in accordance with Standard MIL-STD-106.

3.3.12 In-Phase Linearity. - Percent in-phase linearity error is given by

$$100 \left| \frac{- \} \quad E \; .}{N' \; (\sim \;) \; ‘ \; o} \right|$$

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where E, E_o, N, No, and N' are as defined in 3.3.6, 3.3.7, and 3.3.8.3.1.

3.3.13 Temperature Rise (Stalled). - Temperature rise at stall (degrees centigrade) is given by

$$(234.5 + t) \frac{\% - \text{'C}}{\text{' c}}$$

where R_{RH} = resistance of windings at equilibrium temperature,
t_c = resistance of windings at starting temperature,
= ambient temperature [degrees centigrade). See 3.3.1.

3.3.14 Tropical Environment. - Tropical environment is defined as conditions of high humidity, high temperature, and spore exposure, which present an opportunity for fungi growth.

3.4 Performance Requirements and Product Characteristics. - The motor -tachometer generator shall meet the following performance require - ments and product characteristics:

3.4.1 Mechanical requirements.

3.4.1.1 End Play. - When tested in accordance with 4.4.4.1, the total end play of the shaft shall be not less than 0.001 inch and not more than 0.005 inch total indicator reading.

3.4.1.2 Radial **Play. -** When tested in accordance with 4.4.4.2, the radial play of the shaft shall not exceed 0.0008 inch total indicator reading.

3.4.1.3 Run-Out. - When tested in accordance with 4.4.4.3, the shaft run out shall not exceed 0.001 inch total indicator reading.

3.4.1.4 Concentricity. - When tested in accordance with 4.4.4.4, *the* shaft axis and the mounting diameter shall be concentric within 0.001 inch total indicator reading.

3.4.1.5 Perpendicularity. - When tested in accordance with 4.4.4.5, the shaft axis and the mounting face shall be perpendicular within 0.001 inch total indicator reading.

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3.4.1.6 Mechanical Noise. - When tested in accordance with 4.4.4.6, the motor-tachometer generator shall be free from abnormal mechanical noise.

3.4.2 Electrical requirements,

3.4,2.1 Insulation Resistance. - When tested in accordance with 4.4.4.7, the insulation resistance between each winding of the motor-tachometer generator and the housing and between each winding and any other winding shall be at least 10 megohms.

3.4.2.2 Direction of Rotation. - When tested in accordance with 4.4.4.8, the direction of rotation shall be counterclockwise when facing the shaft end of the motor-tachometer generator.

3.4.2.3 Phasing. - When tested in accordance with 4.4.4.9, the phase difference between the voltages of terminals P_1 and S_1 of the tachometer shall be zero degrees.

3.4.2.4 Starting Voltage. - When tested in accordance with 4.4.4.10, the starting voltage for both clockwise and counterclockwise directions of rotation shall not exceed 1.5 volts.

3.4.2.5 No-Load Speed. - When tested in accordance with 4.4.4.11, the no load speed for both clockwise and counterclockwise directions of rotation shall not be less than 4800 revolutions per minute.

3.4.2.6 Single Phasing. - When tested in accordance with 4.4.4.12, the rotor shall come abruptly to a standstill from a no-load speed condition when either the control field winding or fixed field winding is de-energized.

3.4.2.7 Stall Torque, - When tested in accordance with 4.4.4.13, the stall torque for both clockwise and counterclockwise directions of rotation shall not be less than 1.45 ounce-inch.

3.4.2.8 Speed at 0.85 Ounce-Inch Torque. - When tested in accordance with 4.4.4.14, the speed with 0.85 ounce-inch torque shall be 2800 plus or minus 280 revolutions per minute for both clockwise and counterclockwise directions of rotation.

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3.4,2.9 Input Power. - When tested in accordance with 4.4.4.15, the power input to each motor-tachometer generator phase shall not exceed 6.6 watts.

3.4.2.10 Zero-speed (null) voltage.

3.4.2 .10.1 Axis Voltage. - When tested in accordance with 4.4.4.16.1, the in-phase axis voltage and the quadrature axis voltage shall not exceed plus or minus 5 millivolts.

3.4.2 .10.2 Position Voltage. - When tested in accordance with 4.4.4.16.1, the in-phase position voltage shall not exceed plus or minus 1 millivolt, and the quadrature position voltage shall not exceed plus or minus 6 millivolts.

3.4.2 .10.3 Harmonics. - When tested in accordance with 4.4.4.16.2, the harmonic null voltage shall not exceed 10 millivolts.

3.4.2.11 **Transformer ratio.**

3.4.2 .11.1 Magnitude. - When tested in accordance with 4.4.4.17.1, the ratio of the in-phase component of the tachometer output voltage to the tachometer energizing voltage shall be 0.03130 plus or minus 0.00063 at 3600 revolutions per minute.

3.4.2 .11.2 Phase Shift. - When tested in accordance with 4.4.4.17.2, the phase shift of the tachometer output voltage with respect to the tachometer energizing voltage shall be zero plus or minus 0.2 degrees at 3600 revolutions per minute.

3.4.2 .11.3 Error Due to Temperature Variation. - When tested in accordance with 4.5.5.1, the maximum variation in transformation ratio shall not exceed plus or minus 0.05 percent over the temperature range of 20 to 70 degrees centigrade.

3.4.2 .11.4 Error Due to Voltage Variation. - When tested in accordance with 4.5.5.2, the variation in transformation ratio with tachometer excitation voltage shall not exceed 0.01 percent per volt over the range of rated tachometer excitation voltage plus and minus 6 percent.

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3.4.2 .11,5 Error Due to Frequency Variation. - When tested in accordance with 4.5.5.3, the average variation in transformation ratio with tachometer excitation frequency shall not exceed 0.01 percent per cycle over the frequency range of 400 plus and minus 20 cycles per second.

3.4.2.12 Linearity. - When tested in accordance with 4.5.6, the in-phase linearity of the tachometer generator shall be zero plus or minus 0.05 percent.

3.4.2.13 Temperature Rise @tailed). - When tested in accordance with 4.5.7, the temperature rise of the control motor-tachometer gener - ator shall not exceed 60 degrees centigrade.

3.4.2.14 Operating Life. - After 1000 hours of operation in accordante with 4.5.8, the motor-tachometer generator shall meet the no-load speed requirement of 3.4.2.5 with 4320 instead of 4800 taken as the minimum required speed in revolutions per minute, and shall meet the stall torque requirement of 3.4.2.7.

3.5 Environmental Requirements. - The motor-tachometer generator shall meet the other applicable requirements of this specification during and after tests defined or specified in 4.5.

3.5.1 Vibration, Operating Condition. - The motor-tachometer generator shall be capable of withstanding the Type I vibration requirement as specified in Standard MIL-STD-167 and shall meet all performance and product characteristic requirements of this specification during and after vibration tests and thereafter show no damage.

3.5.2 Shock, Operating Condition. The motor-tachometer generator shall be capable of withstanding the Type A shock requirements as specified in Specification MIL-S-901 and shall function before and after the shock test and thereafter shall show no damage and meet all performance and product characteristic requirements of this specification.

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3.5.3 Temperature, - The motor-tachometer generator shall be capable of withstanding operational ambient temperatures of between zero degrees Fahrenheit (minus 18 degrees centigrade) and 185 degrees Fahrenheit (85 degrees centigrade) in accordance with Specification MIL-F-18870 and storage temperatures of between minus 40 degrees Fahrenheit (minus 40 degrees centigrade) and 160 degrees Fahrenheit (71 degrees centigrade) in accordance with Condition II of NAVWEPS OS 6341, and thereafter shall show no significant corrosion or damage and shall meet the performance and product characteristics of this specification.

3.5.4 Humidity. - The motor-tachometer generator shall be capable of withstanding operational ambient humidities in accordance with Specification MIL-F-18870.

3.5.5 Tropical Environment. - The motor-tachometer generator shall **be** capable of withstanding the tropical environment conditions as specified in Specification MIL-E-5272 under "Fungus Resistance Tests" and thereafter shall show no significant fungus growth, corrosion or other damage, and shall meet the performance and product characteristics of this specification.

3.6 Clean-Up. - The assembly shall be thoroughly cleaned of loose metal chips and other foreign matter, Burrs and **sharp edges shall be** removed.

3.7 Workmanship. - The assembly shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of marking, machine screw assemblage, and freedom of parts from burrs and sharp edges.

3.7.1 The standards of workmanship exhibited in any approved preroduction sample, subject to any qualification stated in the Governments otice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by aplicable specifications.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Gages. - The contractor shall provide himself with whatever gages are necessary and adequate to ensure that the material to which this specification applies will meet the requirements shown on the applicable drawings. If so stipulated in the contract, order or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required in the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

4.1.2 Pre-production Sample. - Unless otherwise specified for each contract or order, three motor-tachometer generators shall be manufactured using the methods and procedures proposed for the production lot. These shall be submitted to the Government inspector for pre - production tests at an activity designated by the Bureau of Naval Weapons for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on the pre-production samples shall be at the contractor?s risk. Pre-production samples will be applied as part of the quantity specified by the contract or order.

4.1.3 Periodic Production Sampling, - Periodic production samples shall be selected from units that have satisfactorily passed the acceptance inspections of 4.4. These samples shall be subjected to the tests of 4.5 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

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Step 1. Upon initiation of production, randomly select one (1) unit from each group of ten (10) units. Upon satisfactory compliance of samples selected from three (3) consecutive groups, shift to Step 2.

Step 2. Randomly select one (1) unit from each group of twenty-five (25) units. Upon satisfactory compliance of samples from three (3) consecutive groups, shift to Step 3.

Step 3. Randomly select one (1) unit from each group of fifty (50) units. Continue with Step 3 for the balance of the contract or until a defective unit is found.

When a defective unit is found in any group, acceptance of all units on hand shall ‘be withheld and the contractor shall:

a. Determine the cause and effect.

b. Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

c. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

Return to Step 1 and proceed as indicated when inspection is resumed.

4.2 Classification of Tests. - The inspection and testing of the motor-tachometer generator shall be classified as follows:

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4.2.1 Acceptance Tests. - These tests shall be accomplished on each motor-tachometer generator being submitted for acceptance under contract. These tests are detailed in 4.4.

4.2.2 Pre-production and Periodic Tests. - Pre-production and periodic production tests shall be accomplished on samples selected as specified in 4.1.2, and 4.1.3, which are representative of the production of the motor-tachometer generator after the award of the contract to determine that the production meets the requirements of this specification. These are detailed in 4.5. Acceptance shall be based on no defects in the sample.

4.3 Test Equipment. - The following items of test equipment are required to perform the acceptance tests set forth in this specification:

4.3.1 End Play Tester. - An end play tester or bench lathe capable of securing the motor-tachometer generator in a vertical or horizontal position and mounting a dial indicator.

4.3.2 Dial Indicator. - A dial indicator which can be mounted on the tester specified in 4.3.1, and which measures with an accuracy of 0.0001 inch.

4.3.3 Mechanical loads.

4.3.3.1 Reversing l-Pound Load. - A reversing 1-pound load which can be applied axially to the shaft of the motor-tachometer generator.

4.3.3.2 Reversing l/2-Pound Load. - A reversing l/2-pound load which can be applied radially to the shaft of the motor-tachometer generator.

4.3.3.3 Dynamometer. - A dynamometer capable of applying a maximum 5 ounce-inch torque with a dial calibrated to O. 1 ounce-inch and an accuracy of at least 3 percent. The dynamometer should be capable of speeds up to 10, 000 revolutions per minute.

4.3.4 Power sources.

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4.3.4.1 400-Cycle Motor Power Supply. - A 400 cycle, two-phase power supply having the following characteristics:

(a) Volt-amperes per phase. 25 minimum.
(b) Volts per phase. 115 volts plus or minus 1 percent for motor fixed phase, and continuously adjustable from O to 44 volts minimum for motor control phase.
(c) Phase angle between voltages of fundamental frequency. The control-phase supply voltage shall lead the fixed-phase supply voltage by 90 plus or minus 5 degrees.
(d) Frequency, 400 plus or minus 5 cycles per second.
(e) Waveform. Sinusoidal, with crest factor of 1.41 plus or minus 10 percent, form factor of 1.11 plus or minus 5 percent, and harmonic content not greater than 5 percent of the voltage of fundamental frequency.

4.3.4.2 400-Cycle Generator Power Supply. - A 400-cycle, single-phase supply having the following characteristics:

[a) Volt-amperes. 15 minimum.
(b) Volts. 115 plus or minus 1 percent.
(c) Frequency. 400 cycles per second plus or minus 0.1 percent.
(d) Waveform. Sinusoidal, with harmonic content not greater than O. 1 percent of the voltage of fundamental frequency.

4.3.4.3 60-Cycle Equipment Power Supply. - A 60-cycle power source capable of supplying adequate volt-amperes to power any W-cycle test equipment used in conducting the tests of this specification. The voltage shall be 115 plus or minus 2 volts, the frequency 60 plus or minus 3 cycles per second, and the waveform shall be sinusoidal with harmonic content not greater than 5 percent of the voltage of fundamental frequency.

4.3.5 Adjustable Speed Drive. - A drive which can supply output shaft speeds up to 4200 revolutions per minute in increments of 600 revolutions per minute, and has a stall torque of at least 7 ounce-inches and a speed accuracy of plus or minus O. 005 percent.

4.3.6 Resistor. - A non-inductive resistor having a resistance of 25,000 ohms plus or minus 1 percent and a power rating of 1 watt or greater.

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4.3.7 Meters.

4.3.7.1 Voltmeter. - A multirange alternating-current voltmeter, 1, 3, 10, 30, 100 volts full scale, having a resistance of at least 20,000 ohms per volt and a full scale accuracy of at least plus or minus 2 percent.

4.3.7.2 Ohmmeter. - An ohmmeter capable of measuring resistance to an accuracy of at least 2 percent.

4.3.7.3 Wattmeter. - A wattmeter capable of measuring up to 15 watts which can be used with voltages up to 115 volts and currents up to 1ampere at 400 cycles per second with a full scale accuracy of at least 2 percent.

4.3.8 Electronic instruments.

4.3.8.1 High Potential Tester. - A high potential tester capable of applying 500 and 1000 volts, 60 cycles per second. The output voltage, when turned on or off, shall change at a rate not greater than 75 volts per second.

4.3.8.2 Insulation Resistance Tester. - An insulation resistance tester capable of applying 500 volts direct current, and measuring resistance up to 100 megohms.

4.3.8.3 Phase-Sensitive Voltmeter. - A phase-sensitive voltmeter having the following characteristics:

[a) Capability of reading either in-phase or quadrature component of a 400-cycle per second signal voltage, with polarity indication in accordance with 3.3.8.1.1 and 3.3.8.1.2,

(b) Signal-voltage input impedance not less than 2 megohms shunted by 25 picofarads,

(c) Reference-voltage input impedance not less than 6000 ohms,

(d) Noise and harmonic rejection of 400 decibels minimum,

(e) Sensitivity of 6 millivolts minimum full scale,

(f) Accuracy of plus or minus 2 degrees minimum in tangent of phase angle computed in accordance with 3.3.11,

(g) Center-zero scale.

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4.3.8.4 Stroboscope. - A stroboscope capable of flashing at rates up to 10,000 flashes per second, and having an accuracy of at least 2 percent of indicated flashing rate.

4.3.8.5 Vacuum Tube Voltmeter. - Input impedance at least 500,000 ohms, lowest full scale reading not greater than 10 millivolts, accuracy at least 3 percent.

4.3.9 Filter. - A notch filter which has a transmission that is down at least 40 decibels at 400 cycles per second, down not more than 3 decibels at 200 and 800 cycles, and down not more than 1 decibel at frequencies outside the 133 to 1200 cycle per second band.

4,3.10 Ninety-Degree Phase Shifter. - A network which will shift a 400-cycle per second sine wave by 90 plus or minus 1 degree when it is driven from a zero impedance source and has the precision ratio transformer of 4.3.11.3 as a load.

4.3.11 Transformers

4.3.11.1 Step-Down Isolation Transformers. - Two step-down isolation transformers which step down 115 volts, 400 cycles per second to 6 volts plus or minus 1 volt. These transformers should be rated at 10 watts minimum.

4.3.11.2 Isolation Transformer. - An audio-frequency isolation transformer having a turns ratio of approximately 1 to 1.

4.3.11.3 Precision Ratio Transformers. - Two precision ratio transformers with continuous resolution, and an accuracy of plus or minus 0.001 percent from 30 to 1000 cycles per second. This transformer should be rated at 150 or more volts maximum input voltage, and have an input inductance of approximately 500 henrys.

4.3.12 Heat Sink. - A heat sink as shown in figure 6 on which the motor-tachometer generator can be mounted.

4.4 Acceptance tests.

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4.4.1 The contractor shall conduct acceptance tests as specified in 4.2.1 to assure that the control motor-tachometer generator is in compliance with the requirements of 3.2, 3.4, 3.6, and 3.7.

4.4.2 Test conditions.

4.4.2.1 Temperature. - Room ambient 65 degrees Fahrenheit to 95 degrees Fahrenheit (18 degrees centigrade to 35 degrees centigrade).

4.4.2.2 Altitude. - Normal ground atmospheric pressure.

4.4.2.3 Vibration. -None.

4.4.2.4 Humidity, - Room ambient to 95 percent relative maximum.

4.4.3 Test and Inspection Equipment and Facilities. - The manufacturer shall furnish and maintain all necessary test equipment, facilities, and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.4.4 Test Procedure. - The detailed test procedure listed in the following paragraphs shall be followed.

4.4.4.1 End Play. - With the motor-tachometer generator mounted horizontally in the tester of 4.3.1 and the housing held stationary, the reversing l-pound load of 4.3.3.1 shall be applied axially to the shaft with the indicator of 4.3.2 mounted to measure axial movement of the shaft. The total indicator reading shall be observed to determine compliance with the requirements of 3.4.1.1.

4.4.4.2 Radial Play. - With the motor tachometer generator mounted horizontally in the tester of 4.3.1 and the housing held stationary, the reversing 1/2 -pound load of 4.3. 3.2 shall be applied radially to a point on the shaft not more than 1/8 inch from the bearing with the indicator of 4. 3.2 mounted to contact the shaft nor more than 1/4 inch from the bearing. The total indicator reading shall be observed to determine compliance with the requirements of 3.4.1.2.

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4.4.4.3 Run-Out. - With the motor-tachometer generator mounted in the tester of 4.3.1 and the housing held stationary, the shaft shall be rotated with the indicator of 4.3.2 mounted to contact the shaft within 1/16 inch of its end. The total indicator reading shall be observed to determine compliance with the requirements of 3.4.1.3.

4.4.4.4 Concentricity. - With the motor-tachometer generator mounted vertically in the tester of 4. 3,1 and the shaft held stationary, the housing shall be rotated by hand with the indicator of 4.3.2 mounted to contact the mounting diameter of the unit. The total indicator reading shall be observed to determine compliance with the requirements of 3.4.1.4.

4.4.4.5 Perpendicularity. - With the motor-tachometer mounted vertically in the tester of 4.3.1 and the shaft held stationary, the housing shall be rotated by hand with the indicator of 4.3.2 mounted to contact the mounting face not more than O. 062 inch from its outside diameter. The total " indicator reading shall be observed to determine compliance with the requirements of 3.4.1.50

4.4.4.6 Mechanical Noise. - With the motor-tachometer generator connected as shown in figure 1, and the input voltage adjusted to give a voltmeter reading of 40 volts, the unit shall be checked for freedom from abnormal mechanical noise such as knocking or intermittent scraping sounds for both directions of shaft rotation in the horizontal, vertical - shaft up, and vertical - shaft down positions to determine compliance with the requirements of 3.4.1.6.

4.4.4.7 **Insulation resistance.**

4.4.4.7.1 Application of High Potential. - With no electrical connections made to the motor-tachometer generator, the following voltages shall be applied for 15 plus or minus 5 seconds using the high potential tester of 4.3.8.1 as indicated below.

4.4.4 .7.1.1 500 volts.

(a) housing to green lead
(b) housing to blue lead
(c) green lead to blue lead
(d) housing to terminal S_1

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4.4.4 .7.1.2 1000 volts. “

(a) housing to black lead
(b) housing to either red lead

4.4.4.7.2 Immediately after the motor-tachometer generator has been subjected to the above voltages, the resistance between each winding and any other winding, and between each winding and the housing shall be checked using the insulation resistance tester of 4.3.8.2 to determine compliance with the requirements of 3.4.2.1.

Note: The initial test by the vendor is to be made at the amplitude shown above. Any subsequent application of high potential shall be at 80 per cent of the voltages specified above. Excessive high potential testing should be avoided.

4.4.4.8 Direction of Rotation. - With the motor-tachometer generator concerned as shown in figure 1, switch S_1 in the position shown in figure 1, and the input voltage adjusted to give a voltmeter reading of 40 volts, the direction of shaft rotation shall be observed to determine compliance with the requirements of 3.402.2.

4.4.4.9 Phasing. - With the motor-tachometer generator connected as shown in figure 1, the input voltage adjusted to give a voltmeter reading of 40 volts, and switch S reversed from the position shown, the phase-sensitive voltmeter of 4.3.8.3 shall be used to measure the phase difference between terminals P_1 and S_1 of the tachometer to determine compliance with the requirements of 3.4.2.3.

Note: Test Conditions for 4.4.4.10 Through 4.4,4.17. - In addition to the test conditions specified in 4.4.2, the following conditions shall pertain for the tests of 4.4,4.10 through 4.4.4.17.

1. **Mounting. - The** motor-tachometer generator shall be mounted **on the heat sink specified in** 4.3.13, and shielded from room air currents.

2. Ambient atmosphere.

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(a) Temperature. 25 plus or minus 5 degrees centigrade,
(b) Pressure. 30 plus or minus 2 inches of mercury,
(c) Humidity. Room ambient to 75 percent relative maximum.

3. Temperature Stabilization. - Before conducting any of tests 4.4.4.10 through 4.4.4.17, the motor-tachometer generator shall be given a warm-up period of one hour, connected as shown in figure 1, with switches S and S_4 closed and switch S open. During the warm-up period and thereafter, room ambient temperature shall not vary more than plus or minus 5 degrees centigrade. Switch S shall remain closed during the tests of 4.4.4.10 through 4.4.4.17 unless otherwise specified.

4.4.4.10 Starting Voltage. - With the motor-tachometer generator connected as shown in figure 1, the input voltage shall be increased slowly from zero until the shaft starts to rotate smoothly; the reading of the voltmeter is the starting voltage. The starting voltage shall be checked for at least 3 starts in each direction of shaft rotation to determine compliance with the requirements of 3.4.2.4. See 3.3.2.

4.4,4.11 No-Load Speed. - With the motor-tachometer generator connected as shown in figure 1, and the input voltage adjusted to give a voltmeter reading of 40 volts, the stroboscope of 4.3.8.4 shall be used to check the no-load speed in each direction of shaft rotation to determine compliance with the requirements of 3.4.2.5. See 3.303.

4.4.4.12 Single Phasing. - With the motor-tachometer generator connected as shown in figure 1, the input voltage adjusted to give a voltmeter reading of 40 volts, and the tachometer excitation voltage disconnected, switch S_2 shall be opened after the unit has been running for at least 5 minutes. This procedure shall be repeated, **opening switch S_3 instead of switch S_2. In each case, operation of the unit when the switch is opened shall be observed to determine compliance with the requirements of 3.4.2.6. See 3.3.4.**

4.4.4.13 Stall Torque. - With the motor-tachometer generator connected as shown in figure 1, and the input voltage adjusted to give a voltmeter reading of 40 volts, the dynamometer of 4.3.3.3 shall be used to measure stall torque for both positions of switch S_1 to determine compliance with the requirements of 3.4.2.7. See 3.3.5.

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4.4.4.14 Speed at 0.85 Ounce-Inch Torque. - With the motor-tachometer generator connected as shown in figure 1, and the input voltage adjusted to give a voltmeter reading of 40 volts, the dynamometer of 4.3.3.3 shall be used to obtain a torque of O. 85 ounce inches and the speed shall be measured for each direction of rotation, using the stroboscope of 4.3.8.4 to determine compliance with the requirements of 3.4.2.8.

4.4.4.15 Input Power. - With the motor-tachometer generator connected as shown in figure 1, and the input voltage adjusted to give a voltmeter reading of 40 volts, the wattmeter of 4.3.7.3 shall be used to measure the power input to the main phase and to the control phase of the motor with the shaft locked by means of the dynamometer of 4.3.3.3 to determine compliance with the requirements of 3.4.2.9.

4.4.4.16 Zero speed (null) voltage.

4.4.4 .16.1 Axis Voltage and Position Voltage. - With the motor-tachometer generator connected as shown in figure 1, and with switch S_3 open, the shaft of the unit shall be rotated slowly by hand, the maximum and minimum values of both the in-phase and quadrature components of the voltage between terminals S_1 and S_2 of the tachometer measured using the phase-sensitive voltmeter, and the in-phase and quadrature axis and position voltages computed to determine compliance with the requirements of 3.4.2 .10.1 and 3.4.2.10.2. See 3.3.8.2.1, 3.3.8.2.2, and figures 4 and 5.

4.4,4 .16.2 Harmonics. - With the motor-tachometer generator connected as shown in figure 1, and switch S_3 open, the shaft of the unit shall be rotated slowly by hand and the maximum value of the harmonics component of the voltage between terminals S_1 and S_2 measured using the circuit of figure 3 to determine compliance with the requirements of 3.4.2.10.3. See 3.3.8.2.3.

4.4.4.17 Transformation Ratio. - With the motor-tachometer generator connected as shown in figure 1, with switch S_3 open, and the unit driven at 3600 revolutions per minute by the adjustable speed drive of 4,3.14, the in-phase and quadrature components of the voltage between terminals S_1 and S_2 of the tachometer shall be measured for both directions of rotation using the circuit figure 2, and recorded.

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4.4.4 .17.1 Magnitude. - Using the data obtained in 4.4.4.17, E (see 3.3.8.3. 1) shall be computed, taking N as 3600 revolutions per minute, and the magnitude of the transformation ratio shall be computed as the ratio of E to the voltage between terminals P and P of the tachometer to determine compliance with the requirement s of 3.4.11.1.

4.4.4 .17.2 Phase Shift. - Using the data obtained in 4.4.4.17, Q. (see 3.3.8.3. 2) shall be computed. Using Q and E , the phase shift shall be computed, using the formula of 3.3 .ql to determine compliance with the requirements of 3.4.11.2.

4.4.5 Preparation for Delivery. - Preparation for delivery of the' motor-tachometer generator shall conform to Section 5 of this specification.

4.5 Pre-production and periodic production sample tests.

4.5.1 Vibration, Operating Condition. - The motor-tachometer generator rigidly mounted in a suitable vibration fixture, shall be subjected to vibration tests so as to meet the requirements specified in 3.5.1, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.2 Shock, Operating Condition. - The motor-tachometer generator rigidly mounted to a suitable shock fixture, shall be subjected to shock tests so as to meet the requirements specified in 3.5.2, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.3 Temperature and Humidity. - The motor-tachometer generator shall be subjected to temperature and humidity tests so as to meet the requirements specified in 3.5.3, and 3.5.4, and thereafter shall meet the performance requirements and product characteristics of this specification.

4.5.4 Tropical Environment. - The motor-tachometer generator shall be exposed to fungus tests so as to meet the requirements specified in 3.5.5 and thereafter shall meet all the performance requirements and product characteristics of this specification.

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Note: Test Conditions for 4.5,5 Through 4.5.8. - In addition to the test conditions specified in 4.4.2, the conditions specified in the note between 4.4.4.9 and 4.4.4.10 shall pertain for the tests of 4,5.5 through 4.5.8, except as noted.

4.5.5 Transformation ratio.

4. 5.5.1 Error Due to Temperature Variation. - The magnitude of the transformation ratio shall be measured at ambient temperature of 20, 25, 30, 40, 50, 60, and 70 plus or minus 2 degrees centigrade, using the method of 4.4.4,17.1, and allowing a warm-up period of one hour when going from one temperature to the next. The measured transformation ratios shall be plotted versus temperature on linear-coordinate graph paper, a smooth curve shall be drawn through the points, and the maximum deviation of the curve from the value of transmission ratio measured at 25 plus or minus 2 degrees centigrade shall be noted and multiplied by 100 to determine compliance with the re - quirements of 3.4.2.11.3.

4.5.5.2 Error Due to Voltage Variation, - The magnitude of the transformation ratio shall be measured at tachometer excitation voltages of 108.1, 115.0, and 121.9 volts, using the method of 4.4.4.17.1. The difference of the measured transformation ratios for 108,1 and 115.0 volts, and for 115.0 and 121.9 volts shall be computed and the difference having the greatest absolute value shall be divided by 6.9 and multiplied by 100 to determine compliance with the requirements of 3.4.2.11.4.

4.5.5.3 Error Due to Frequency Variation. - The magnitude of the transformation ratio shall be measured at tachometer excitation voltage frequencies of 380, 400, and 420 cycles per second plus or minus 0.5 cycles per second using the method of 4.4.4.17.1. The difference of the measured transformation ratios for 380 and 400 cycles per second, and for 400 and 420 cycles per second shall be computed, and the difference having the greatest absolute value shall be divided by 20 and multiplied by 100 to determine compliance with the requirements of 3.4.2.11.5.

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4.5.6 Linearity. - With the motor-tachometer generator connected as shown in figure 1, and switch S_3 open, the in-phase output voltage between terminals S_1 and S shall be measured and recorded for speeds of 600, 1200, 1800, 2400, 3000, 3600, and 4200 revolutions per minute for both directions of rotation, using the adjustable speed drive of 4.3.14 and the circuit of figure 2. Using this data, the in-phase linearity shall be computed to determine compliance with the requirements of 3.4.2.12. See 3.3,12. N shall be taken as 3600 revolutions per minute, and N' as 1.2 No. ^{0}

4.5.7 Temperature Rise (Stalled). - With the motor-tachometer generator connected as shown in figure 1, the input voltage adjusted to give a voltmeter reading of 40 volts, and the shaft locked, the unit shall be allowed to remain in this condition for one hour to allow the temperature of the unit to stabilize. At the end of the one-hour temperature-stabilization period, the resistance between tachometer terminals S and S shall be measured (with the 25, 000-ohm resistor disconnected) using the ohmmeter of 403,7,2. The motor-tachometer generator shall then be de-energized and allowed to cool for 1-1/2 hours. The resistance between terminals S_1 and S shall again be measured, and the temperature rise shall be computed to determine compliance with the requirements of 3.4.2.13. See 3.3.13.

4.5.8- Operating Life. - With the motor-tachometer generator connected as shown in figure 1, the input voltage adjusted to give a voltmeter reading of 40 volts; and the shaft free, the unit shall be operated 400 hours in the vertical-shaft up position, then 300 hours in the vertical-shaft down position, and finally 300 hours in the horizontal position. At the end of the 1000 hours of operation, the no-load speed and stall torque shall be measured to determine compliance with the requirements of 3.4.2:14. See 3.4.2.5 and 3.4.2.7.

4.6 Failure of the motor-tachometer generator to meet any of the requirements and tests of this specification shall be considered cause for rejection.

5. PREPARATION FOR DELIVERY

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5.1 Preservation and Packaging. - Preservation and packaging shall be level A, B, or C, as specified (see 6.2).

5.1. 1 Level A

5.1. 1.1 Cleaning. - The motor-tachometer generator shall be cleaned in accordance with method C-1 of Specification MIL-P-II 6.

5.1.1.2 **Drying. - Unless otherwise specified in the contract or order, the selection of the drying procedure to be used shall be at the option of the contractor, provided the procedure selected will not cause damage to the motor-tachometer.**

5.1.1.3 Preservation Application. - None required.

5.1.1.4 Unit Packaging. - The motor-tachometer generator shall be individually packaged before packing to conform to the conditions set forth in method IA-5 of Specification MIL-P-116.

5.1.1.5 **Cushioning. - The unit package and pack shall incorporate sufficient cushioning material, bracing or other adequate shock absorbing devices to prevent damage during shipment and handling.** At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent. The pH range of the material shall be between 6.0 and 8,0.

5.1.2 Level B.

5.1.2.1 Cleaning. - Cleaning of the motortachometer generator shall conform to the requirements of 5.1.1.1.

5.1.2.2 Drying. - Drying of the motor-tachometer generator shall conform to the requirements of 5.1.1.2.

5.1.2,3 Preservation Application. - None required.

5.1.2.4 Unit Packaging. - The packaging of the motor-tachometer generator shall conform to the requirements of 5,1.1.4. “

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5.1.3 Level C. - Cleaning, drying, preservation, and packaging shall be in accordance with the manufacturer's commercial practice, when adequate.

5.2 Packing.

5.2.1 Level A. - Ten motor-tachometer generators, packaged as defined in 5.1, shall be packed in an overseas type wood box conforming to Specification PPP-B-621, Class 2, Style 4, Type 3. The motor-tachometer generators shall be cushioned, blocked, and braced to insure adequate protection during shipment, multiple handling and indeterminate storage.

5.2.2 Level B. - Ten motor-tachometer generators, packaged as defined in 5.1, shall be packed in a domestic type wood box conforming to Specification **PPP-B-621, Class 1, Style 4, Type 3. The motor-tachometer generators shall be cushioned, blocked, and braced to assure adequate protection during shipment, multiple handling and covered storage.**

5.2.3 Level C. - The motor-tachometer generators, when packaged as defined in 5.1, shall be packed to afford protection against damage during direct shipment from supply source to the first receiving activity for immediate use. Containers shall be in accordance with the Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation at the lowest rate.

5.3 Marking. - In addition to any special marking required by the contract or order, all unit packaging and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129.

6. NOTES

6.1 Intended **Use.** - The motor-tachometer generators are intended to be used in the Angle Solvers Mark 18 Mods 2 and 3 and the Fire Control System Mark 112 or other suitable application.

6.2 Ordering Data. - Procurement documents should specify the following:

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(a) The specification title, number and date of this specification,

(b) State whether the commodities shall be afforded preservation and packaging in accordance with level A, B, or C (see 5.1.1, 5.1.2, and 5.1.3),

(c) State whether the commodities shall be afforded packing in accordance with level A, B, or C (see 5.2.1, 5.2.2, and 5.2. 3),

(d) The exceptions to this publication, applicable drawings, and other documents.

6.3 Criteria for use of levels of preservation, packaging and packing.

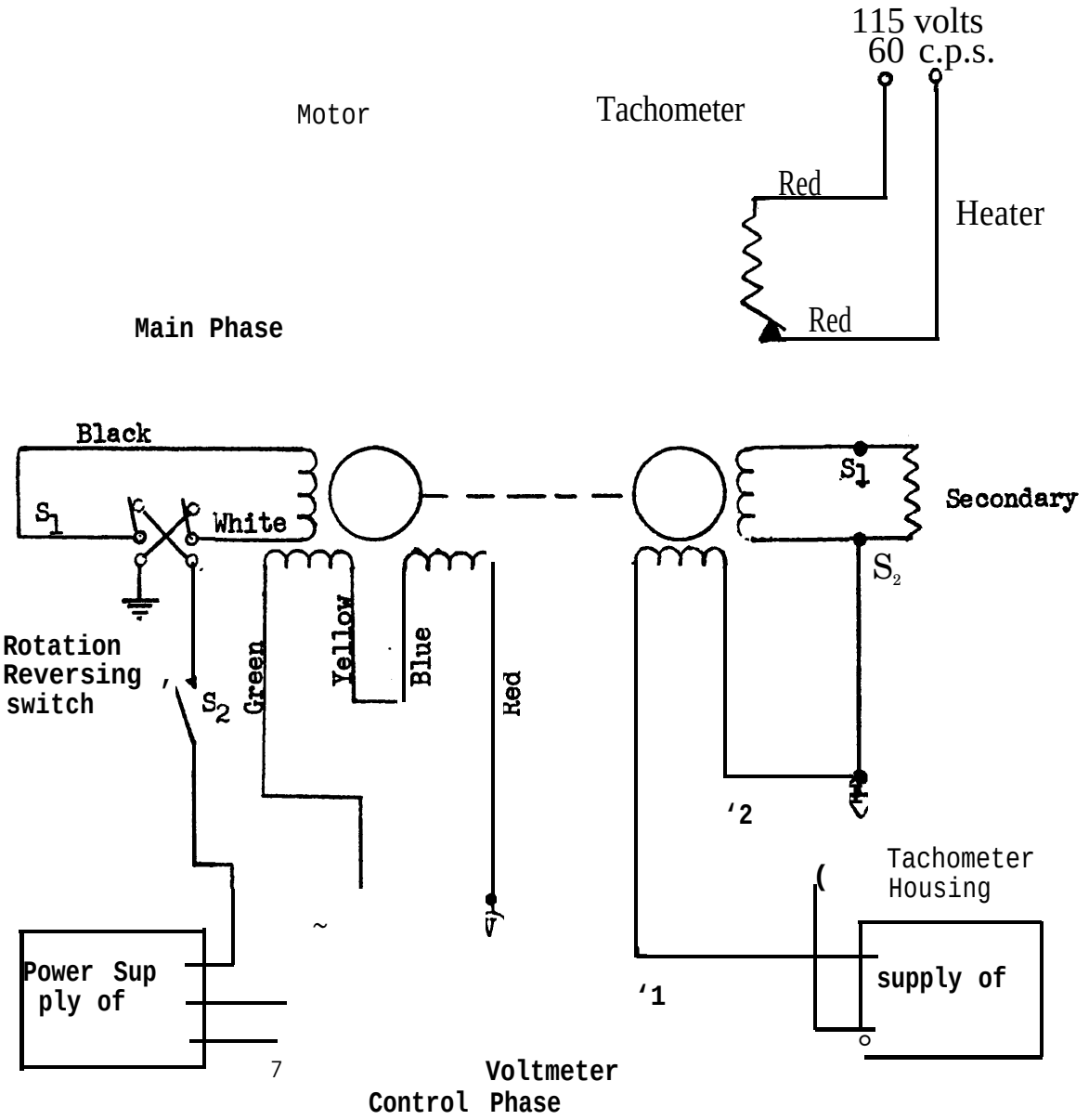
6.3.1 Preservation, packaging and packing levels.

6.3.1.1 Level A. - The degree of preservation, packaging and packing used for this level shall afford adequate protection against corrosion, deterioration and physical damage during handling, shipment, indeterminate storage and world-wide redistribution.

6.3.1.2 Level B. - The degree of preservation, packaging and packing used for this level shall afford adequate protection against known conditions which are less hazardous than Level A is designed to meet. This requires a higher degree of protection than afforded Level C preservation and packaging. The use of Level B should be based on firmly established knowledge of the shipment and storage conditions to be encountered and a determination that a monetary saving will result. The packing shall afford adequate protection against damage during multiple shipment. The packs must be shipped and handled under cover and stored in warehouses or other structures affording equivalent protection from weather.

6.3.1.3 Level C. - The degree of preservation, packaging and packing used for this level shall afford adequate protection against corrosion, deterioration and physical damage during direct shipment from the supply source to the first receiving activity for immediate use. The preservation and packaging may conform to the supplier's commercial practice when such meets the requirements of this level. The packing, in general, will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

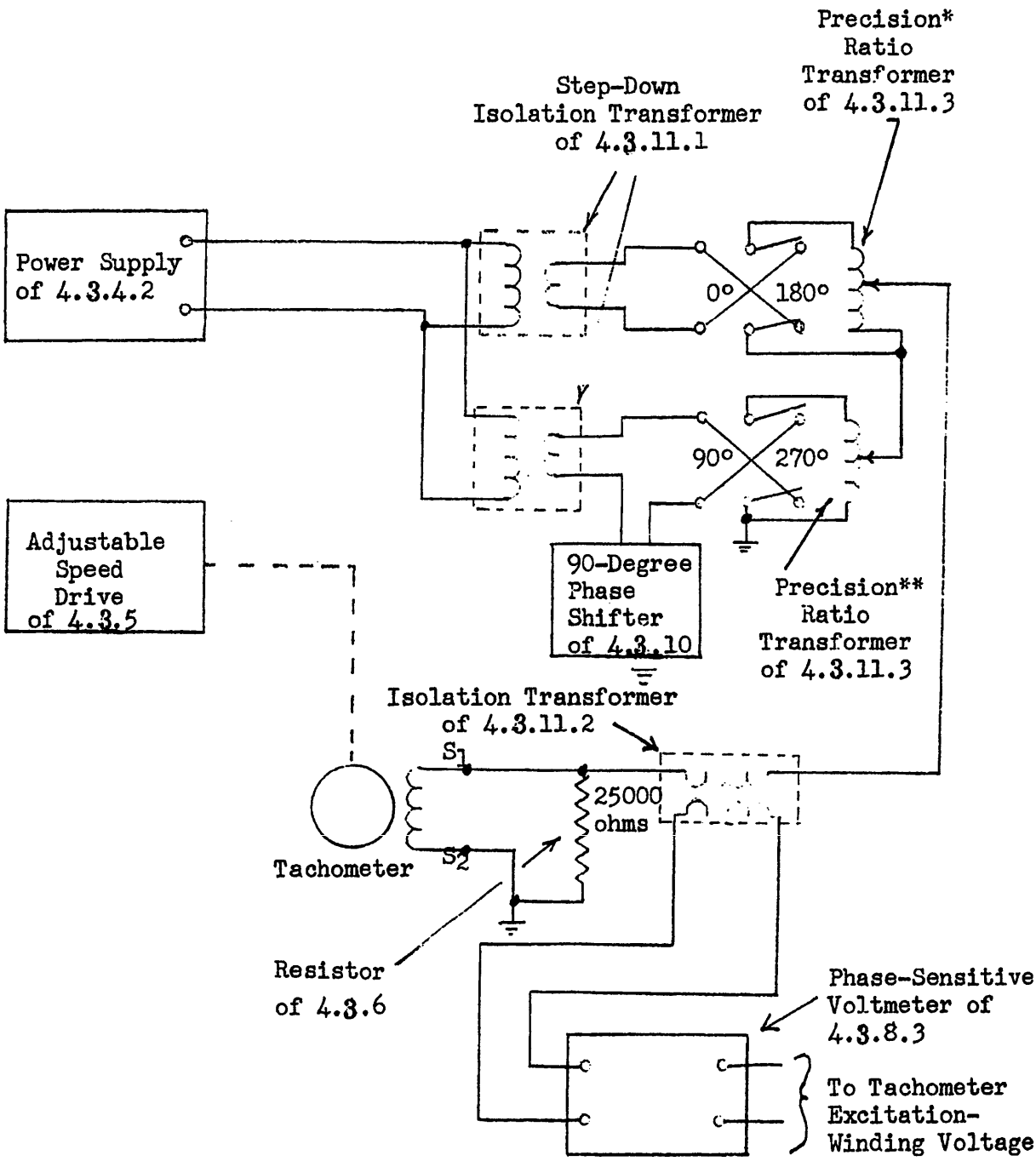
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Note: If the power supply of *4.3.4.2* is used to energize both the motor fixed phase and the tachometer excitation winding (primary), the motor white-to-black voltage shall be in phase "with the tachometer P2-to_Pl voltage.

Figure 1. Test Circuit

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* adjust for in-phase voltage null
** adjust for quadrature voltage null

Figure 2. Circuit of Measurement of Transformation Ratio and Linearity

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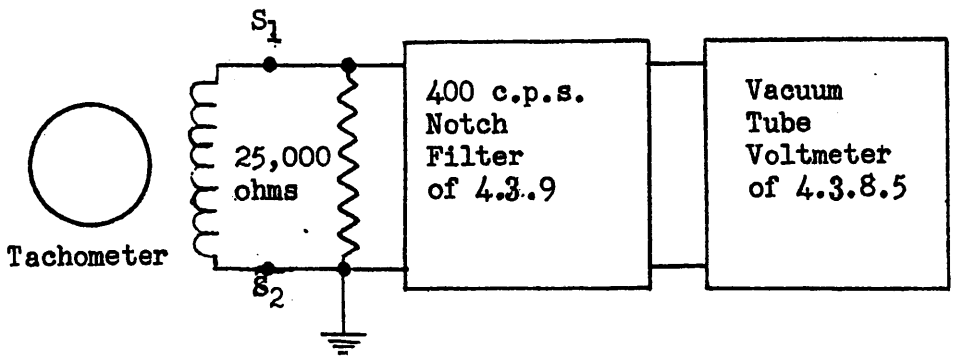


Figure 3. Circuit for Measurement of Zero-Speed Harmonics

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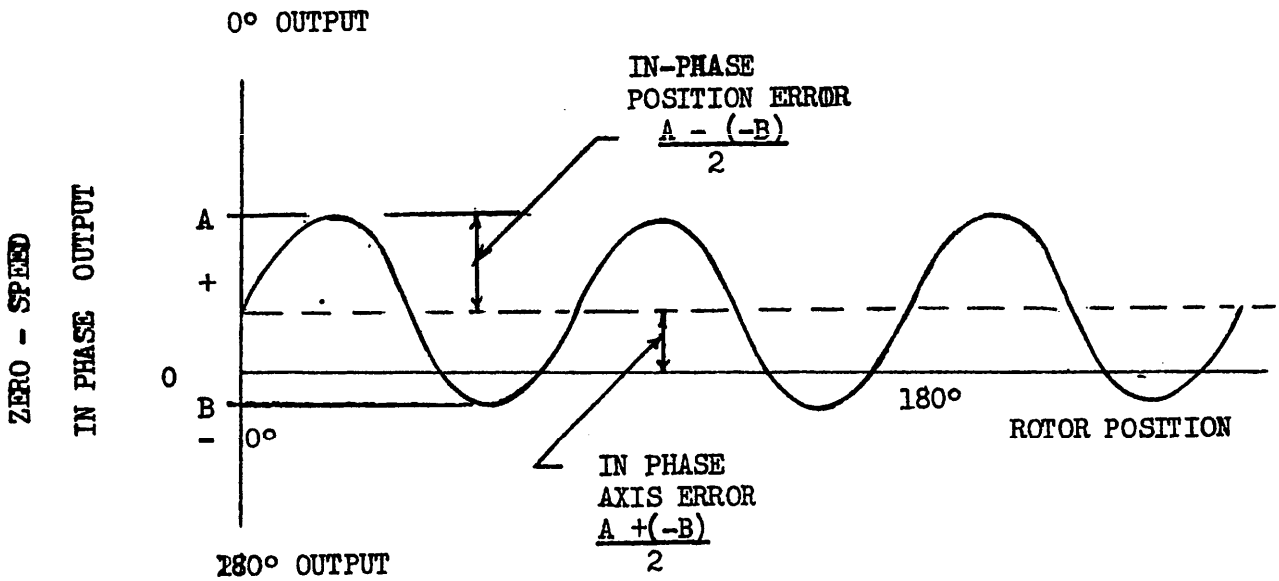


Figure 4. In-Phase Zero-Speed Tachometer Output Voltage versus Rotor Position

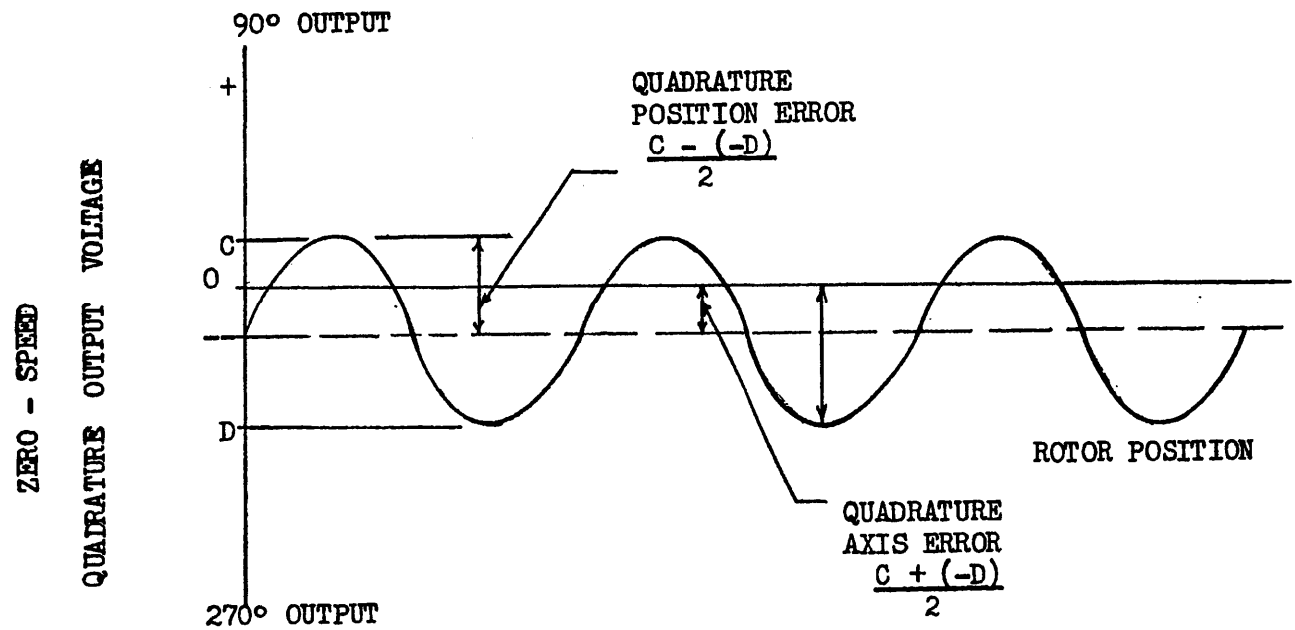
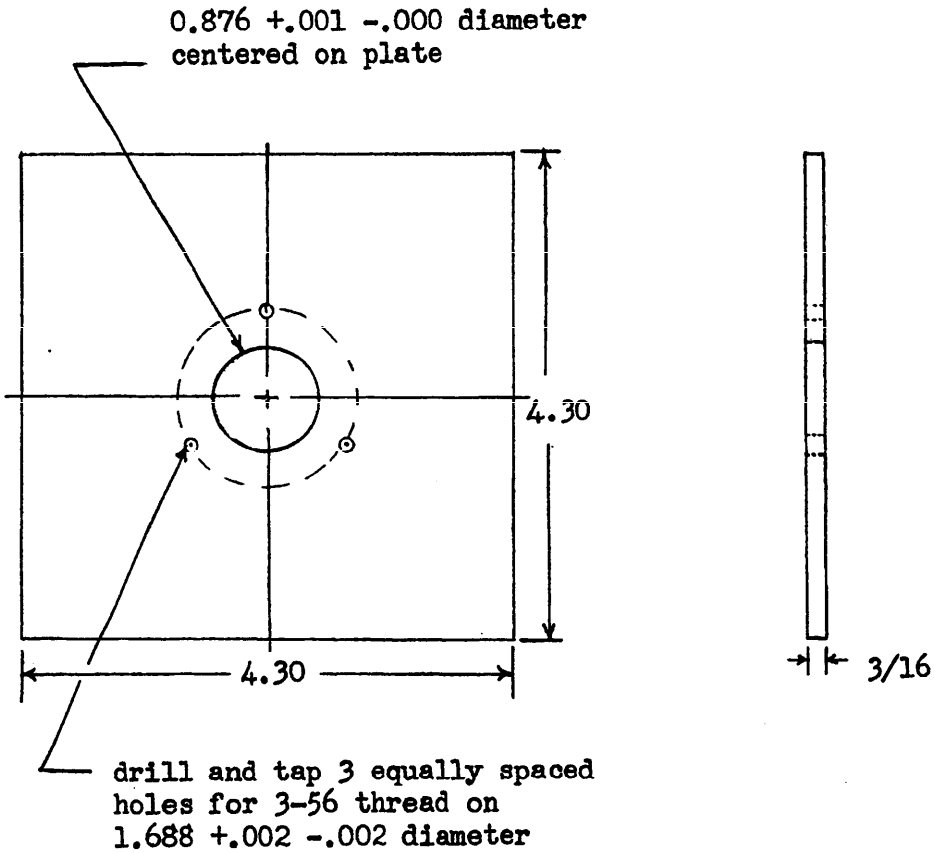


Figure 5. Quadrature Zero-Speed Tachometer Output Voltage versus Rotor Position

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Material. Aluminum, temper T3 flat plate, Specification QQ-A-355.

Finish. Black anodized.

Figure 6. Heat Sink

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-M-23017(Wep) MOTOR-TACHOMETER GENERATOR, TEMPERATURE COMPENSATED

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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